CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Second Semester of B. Pharm (NEW) Examination April 2016
MA141.1 Advanced Mathematics

Date:12/05/2016(Thursday)**Time**: 10:00 a.m. to 01:00 p.m. **Maximum Marks**: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION I is 30 minutes. Answers to SECTION I must be written during the first 30 minutes of the examination.
3. Answers to SECTION II and SECTION III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat sketches wherever necessary.
6. Use of scientific non-programable calculator is allowed.

SECTION I

Q.1. Darken the circle of the correct/most suitable choice. [20]

1. 57^{th} derivative of function $e^x + 5$ is

   (A) e^x (B) $e^x + 57$ (C) $5e^{5x}$ (D) $5e^{57x}$

2. n^{th} derivative of function $e^{kx}\sin(lx+m)$ is

   (A) $(\sqrt{k^2+m^2})^n e^{kx}\sin(lx+m+n\tan^{-1}(\frac{m}{k}))$

   (B) $(\sqrt{k^2+l^2})^n e^{kx}\sin(lx+m+n\tan^{-1}(\frac{l}{k}))$

   (C) $(\sqrt{k^2+m^2})^n e^{kx}\cos(mx+n+l\tan^{-1}(\frac{m}{k}))$

   (D) $(\sqrt{k^2+n^2})^m e^{kx}\cos(lx+m+n\tan^{-1}(\frac{n}{k}))$

3. The value of $\binom{10}{9}$ is

   (A) 90 (B) 19 (C) 1 (D) 10

4. 11^{th} derivative of function $x^9 - 5x^2 + 7$ is

   (A) 0 (B) $99x^2$ (C) $99xt$ (D) 99

5. Which of the following condition is not necessary for Cauchy's mean value theorem.

   (A) $f(x)$ and $g(x)$ are continuous on $[a,b]$.

   (B) $f(x)$ and $g(x)$ are differentiable on (a,b).

   (C) $g'(x) \neq 0$ for $x \in [a,b]$

   (D) $f(a) = f(b)$ and $g(a) = g(b)$

6. According to Lagrange's mean value theorem, if $f(x)$ is continuous on $[a, b]$ and differentiable on (a, b) then

(A) $f(c) = \frac{f(b)-f(a)}{b-c}$
(B) $f'(c) = \frac{f(b)-f(a)}{b-a}$
(C) $f'(c) = 0$
(D) $f'(c) = b - a$

7. Which of the following is true?

(A) Every Taylor series is Macluarin series
(B) Macluarin series expansion arround 0 is Taylor series
(C) Every Macluarin series is Taylor series.
(D) None of the above

8. Which of the following is Maclurin's series expansion of function $f(x)$?

(A) $f(x) = \sum_{n=0}^{\infty} f^n(0)\frac{x^n}{n!}$
(B) $f(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}$
(C) $f(x) = \sum_{n=0}^{\infty}(-1)^n\frac{x^n}{n!}$
(D) $f(x) = \sum_{n=0}^{\infty} f^n(-1)\frac{x^n}{n!}$

9. The order and degree of differential equation $\frac{d^4y}{dx^4} + y = 0$ are

(A) order=1 and degree=4
(B) order=4 and degree=1
(C) order=4 and degree=4
(D) order=1 and degree=1

10. The solution of equation $\frac{dx}{dy} = 1$ is

(A) $x = y + C$
(B) $x = -y + C$
(C) $x = 1$
(D) $y = 1$

11. The general form of linear equation is

(A) $\frac{dy}{dx} + P(y)y = Q(y)$
(B) $x\frac{dy}{dx} + P(y)y = Q(y)$
(C) $\frac{dy}{dx} + P(x)y = Q(x)$
(D) $\frac{dy}{dx} + P(x)y = Q(x)y$

12. what is the differential equation obtain by $y = a\cos x + b\sin x$?

(A) $\frac{d^2y}{dx^2} - y = 0$
(B) $\frac{d^2y}{dx^2} + y = 0$
(C) $\frac{dy}{dx} - y = 0$
(D) $\frac{dy}{dx} + y = 0$

13. If both roots of auxiliary equation of second order homogenous differential equation are 11, then solution of that equation is

(A) $y = C_1e^{-11x} + C_2e^{11x}$
(B) $y = C_1e^{11x} + C_2e^{-11x}$
(C) $y = e^{11x}(C_1 + C_2)$
(D) $y = e^{11x}(C_1 + xC_2)$

14. Which of the following differential equation is not linear?

(A) $\frac{dy}{dx} - y = e^x$
(B) $\frac{dy}{dx} - y \log x = \frac{x}{y}$
(C) $\frac{dx}{dy} - xy = e^y$
(D) $\frac{dy}{dx} = y$

15. If $K(x)$ is an integrating factor of linear differential equation $\frac{dy}{dx} + P(x)y = Q(x)$, then solution of this equation is

(A) $y(K(x)) = \int(Q(x)K(x))\,dx + C_1$
(B) $y(P(x)) = \int(Q(x)K(x))\,dx + C_1$
(C) $y(Q(x)) = \int(P(x)K(x))\,dx + C_1$
(D) None of above,

where C_1 is arbitrary constant.

16. The Wronskian of two functions $y_1 = \cos x$ and $y_2 = \sin x$ is

(A) 1 (B) 2 (C) $\sin x$ (D) $\cos x$

17. $L\,(5487)$=........

(A) $5487s$ (B) $\frac{5487}{s^2}$ (C) $\frac{5487}{s}$ (D) 5487

18. $L[\sin 3t] = ...$

(A) $\frac{3}{s^2-9}$ (B) $\frac{3}{s^2+9}$ (C) $\frac{s}{s^2+9}$ (D) $\frac{s}{s^2-9}$

19. $L[t^2] = ...$

(A) $\frac{6}{s^2}$ (B) $\frac{6}{s^3}$ (C) $\frac{2}{s^2}$ (D) $\frac{2}{s^3}$

20. $L^{-1}[\frac{1}{99s}] = ...$

(A) $99t$ (B) $\frac{99}{t}$ (C) 99 (D) $\frac{1}{99}$

SECTION II

Q.2. Attempt any **Four** of the following. [20]

(a) (i) Find 5^{th} derivatives of function $y = \cos(7x - 9)$. Stat the result which you used? [03]

(ii) Find 4^{th} derivative of function $y = (5x - 3)^5$. [02]

(b) Verify Cauchy's mean value theorem for $f(x) = x^3$ and $2 - x,\ x \in [0, 2]$. [05]

(c) Solve the initial value problem $\frac{dy}{dx} - 3\frac{y}{x} = x^3 + x^2 + x$ with $y(1) = 1$. [05]

(d) The population of a bacteria is known to increase at a rate proportional to the number of people present at a time t. If the population has doubled in 6 hours, how long it will take to triple? [05]

(e) (i) Find Laplace transform of function $e^{5t} \sin 3t$. [03]

(ii) Find inverse Laplace transform of function $\frac{7}{s^2} - \frac{5}{s^2-25} + \frac{s}{s^2+9}$. [02]

(f) If glucose is fed intravenously at a constant rate, the change in the overall concentration $c(t)$ of glucose in the blood with respect to time may be described by the differential equation [05]

$$\frac{dc}{dt} = \frac{G}{100V} - kc.$$

In this equation, G, V and k are positive constants, G being the rate at which glucose is admitted, in milligrams per minute, and V the volume of blood in the body, in liters (around 5 liters for an adult). The concentration $c(t)$ is measured in milligrams per centimeter. The term $-kc$ is included because the glucose is assumed to be changing continually into other molecules at a rate proportional to its concentration. Solve the equation for $c(t)$ using c_0 to denote $c(0)$.

SECTION III

Q.3. Attempt any **Four** of the following. [20]

(a) Find inverse Laplace transform of function $\frac{1}{(s+1)(s-7)}$. [05]

(b) Expand $f(x) = 3x^2 + 5x - 1$ in powers of $(x-1)$. [05]

(c) Using Leibnits's rule. find n^{th} derivative of function $y = x^2 3^{5x}$. [05]

(d) Using method of variation of parameters solve the differential equation $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = e^{3x}$. [05]

(e) A pot of liquid is put on the stove to boil. The temperature of the liquid reaches 170° F and then the pot is taken off the burner and placed on a counter in the kitchen. The temperature of the air in the kitchen is 76° F. After two minutes the temperature of the liquid in the pot is 123° F. How long before the temperature of the liquid in the pot will be 84° F? [05]

(f) Solve the differential equation $\frac{dy}{dx} - 2\frac{y}{x} = x^2$. [05]

Q.4. Attempt any **Four** of the following. [20]

(a) Solve the differential equation $\frac{dy}{dt} + y = e^t$ with initial condition $y(0) = 0$ using Laplace transform. [05]

(b) Solve the differential equation $2xy\,dx + (x^2 + y^2 + \sin y)dy = 0$. [05]

(c) (i) Solve the differential equation $(D^2 + 2D + 3)y = 0$ [03]

(ii) Solve the differential equation $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 0$ [02]

(d) Let population of insect be decreasing at the rate proportional to its population. If the population has decreased to 25 percent in 10 hours, how long will it take to be half? [05]

(e) Verify Lagrange's mean value theorem for $f(x) = x^2 + 5x + 1,\ x \in [-2, 2]$. [05]

(f) Find 8^{th} derivative of function $y = \frac{1}{(x-7)(x+8)}$. [05]